Probability and Statistics

Hypothesis testing: Nonparametric tests

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Nonparametric tests

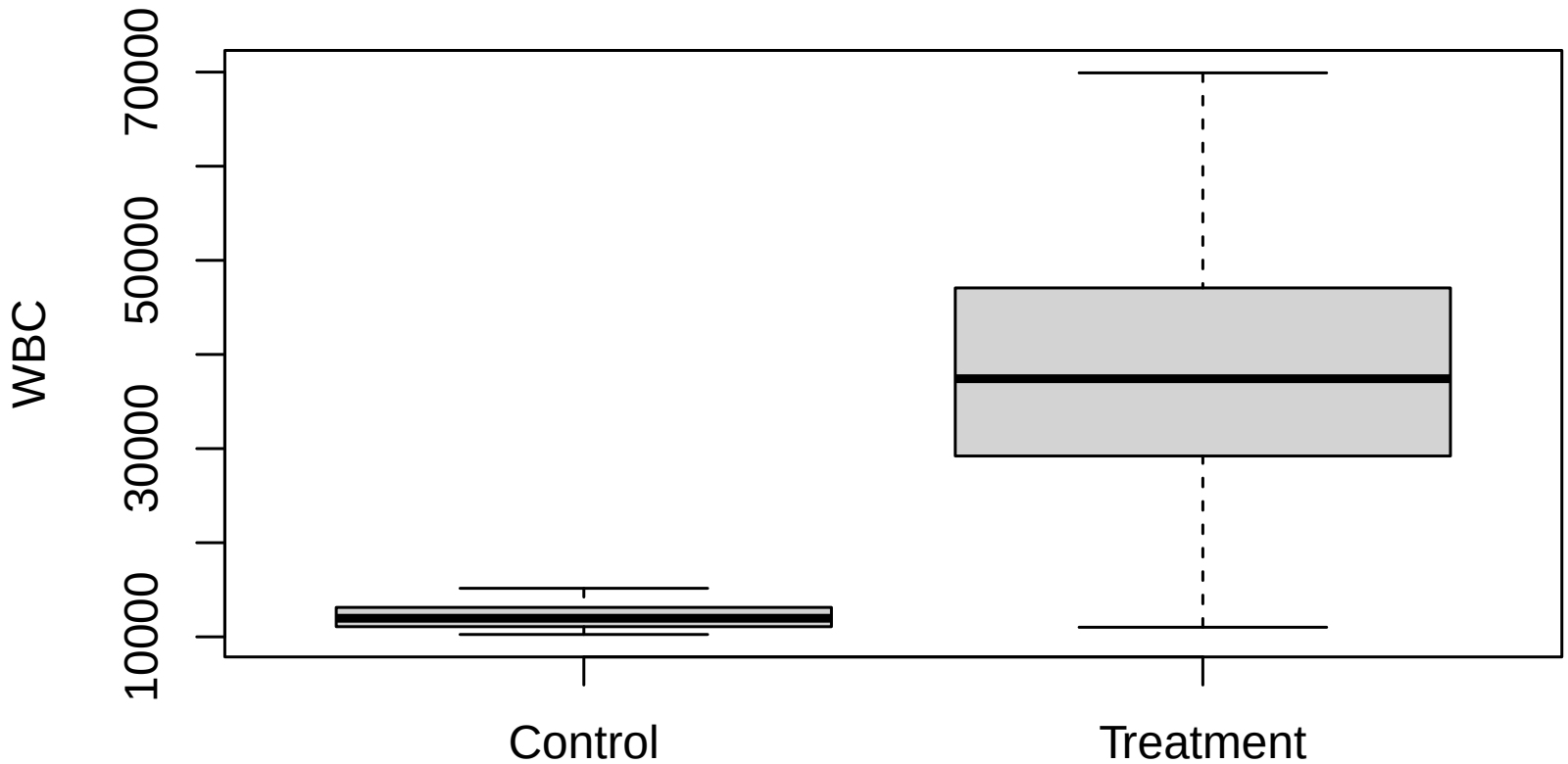
Goal of this lecture

- Develop a testing procedure to compare groups when the data are not necessarily normally distributed.

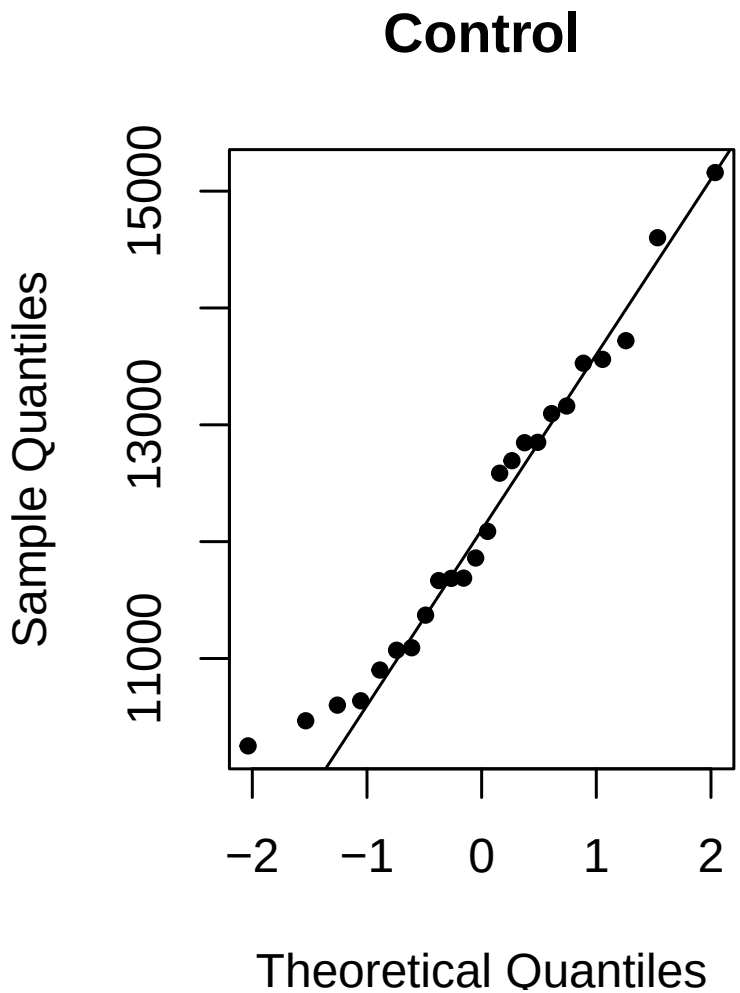
Example: Pertussis in rats

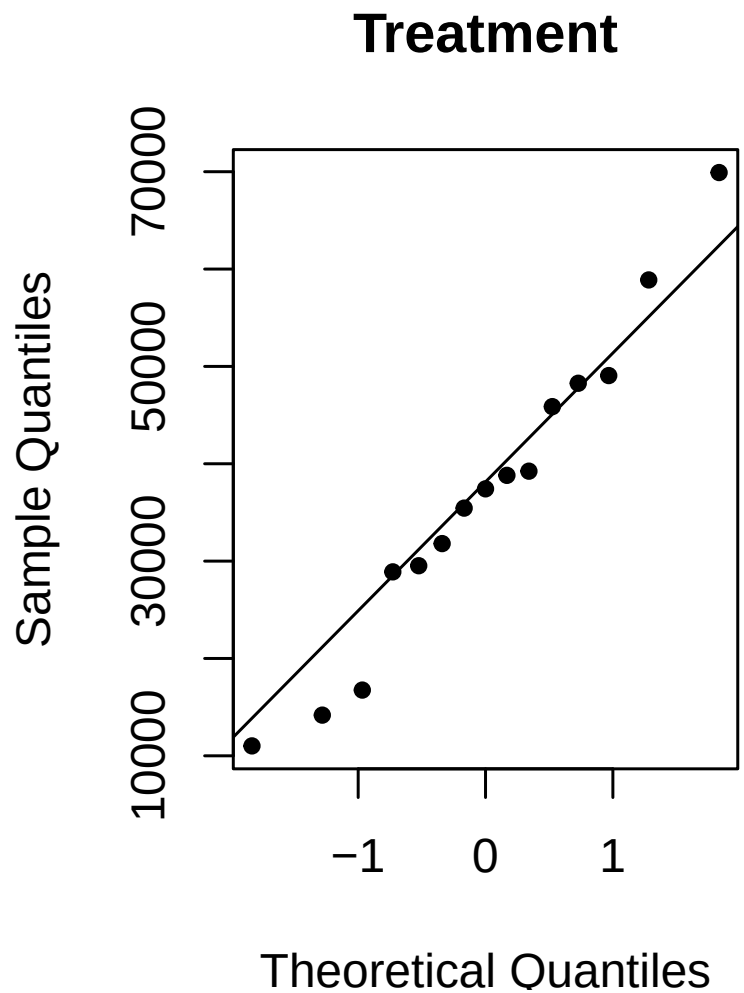
- Pertussis = whooping cough
- Physiological experiment gives concentration of white blood cells (WBC) in blood of
  - 24 control rats
  - 15 rats infected with pertussis in lung
- Research question: is there an association between pertussis and WBC concentration?

Data analysis



QQ-plots for control and treatment





Wilcoxon rank-sum test: setup

- Instead of unpaired t-test, perform **nonparametric test** which makes no assumptions on distribution of observations
- Null hypothesis: WBC has exactly the same distribution in both groups (same shape, center and spread)
  - If we assume both distributions have the same shape, this reduces to: the medians are equal
- Wilcoxon rank-sum test
  - Combine both groups
  - Calculate **rank** of each observation

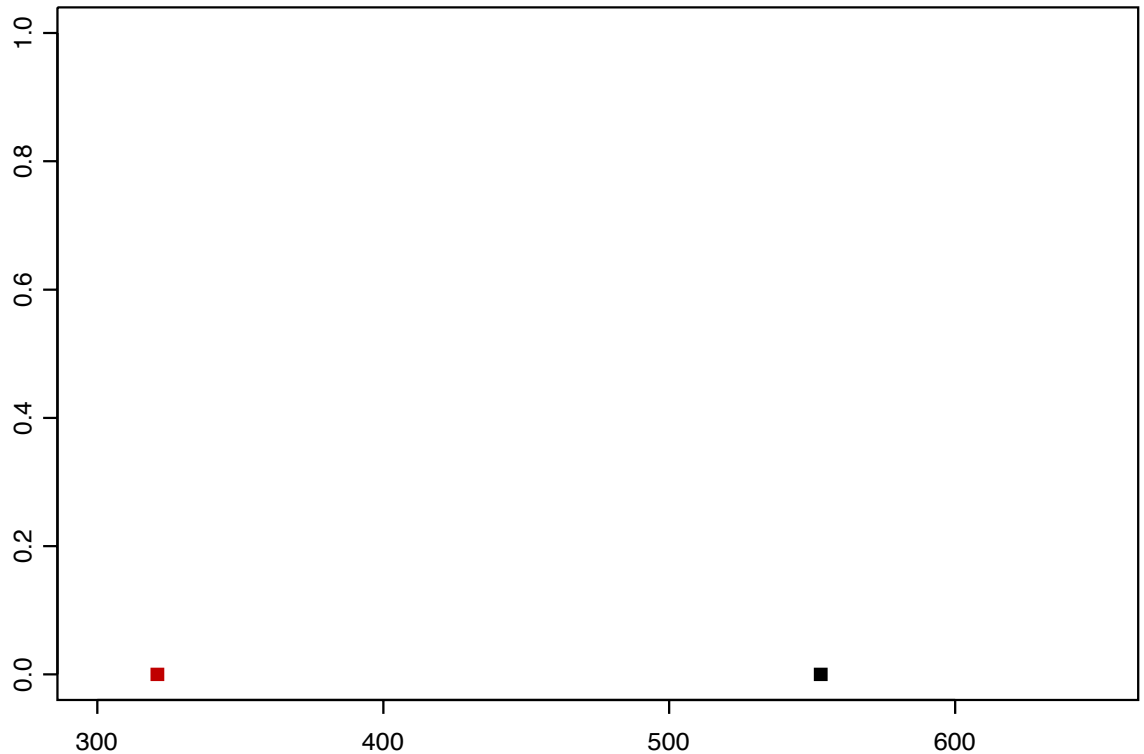
Wilcoxon rank-sum test: test statistic

- Example: Ranks of the measurements 9, 100, 2, 15, 458200 are 2, 4, 1, 3, 5
- Test statistic: rank-sum in control group ($W = 321$)
- Minimum test statistic: the smallest observations are in the control group ($W = 300$)
- Large values ($W \gg 300$) indicate that largest observations stem from control group

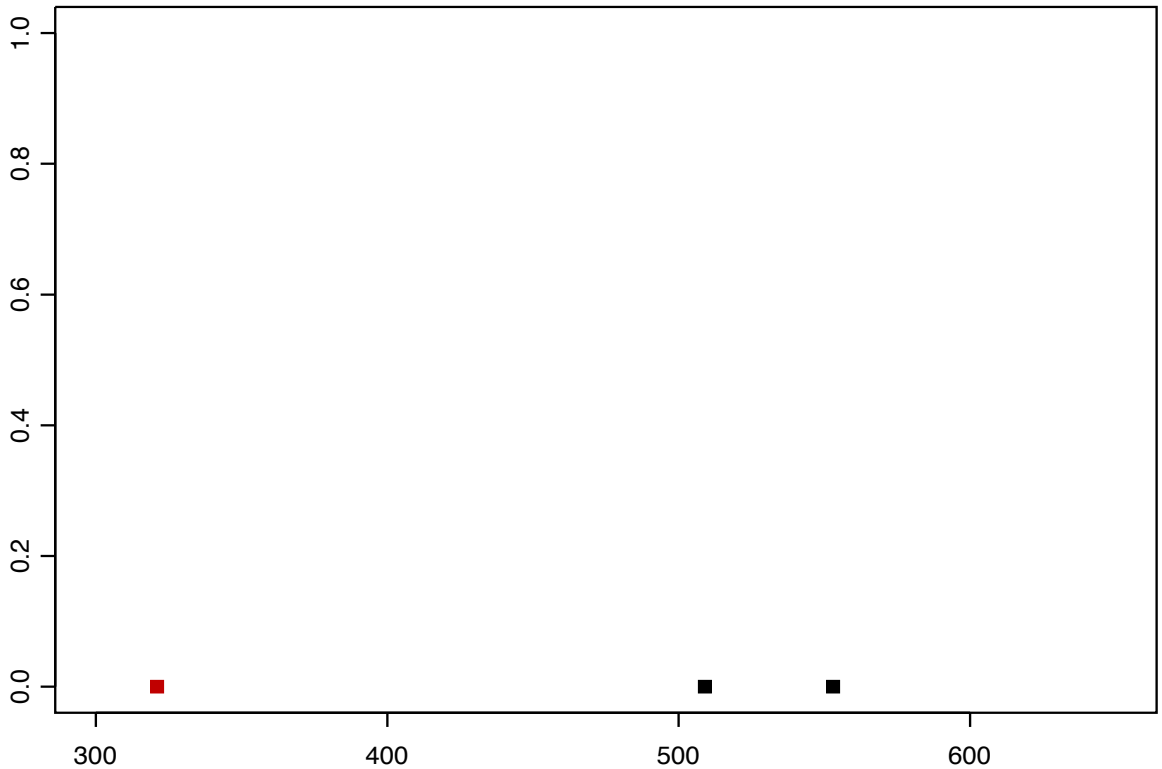
Distribution under null hypothesis

- Under null hypothesis: ranks are randomly divided over both groups
- To determine where test statistic under null hypothesis can be expected, **simulate null hypothesis**:
  - Ranks: $1, 2, ..., 38, 39$
  - Randomly permute ranks, assign first 24 to control group, calculate their sum
- We obtain 553
- This number was obtained under null hypothesis
- *Which null hypothesis*?

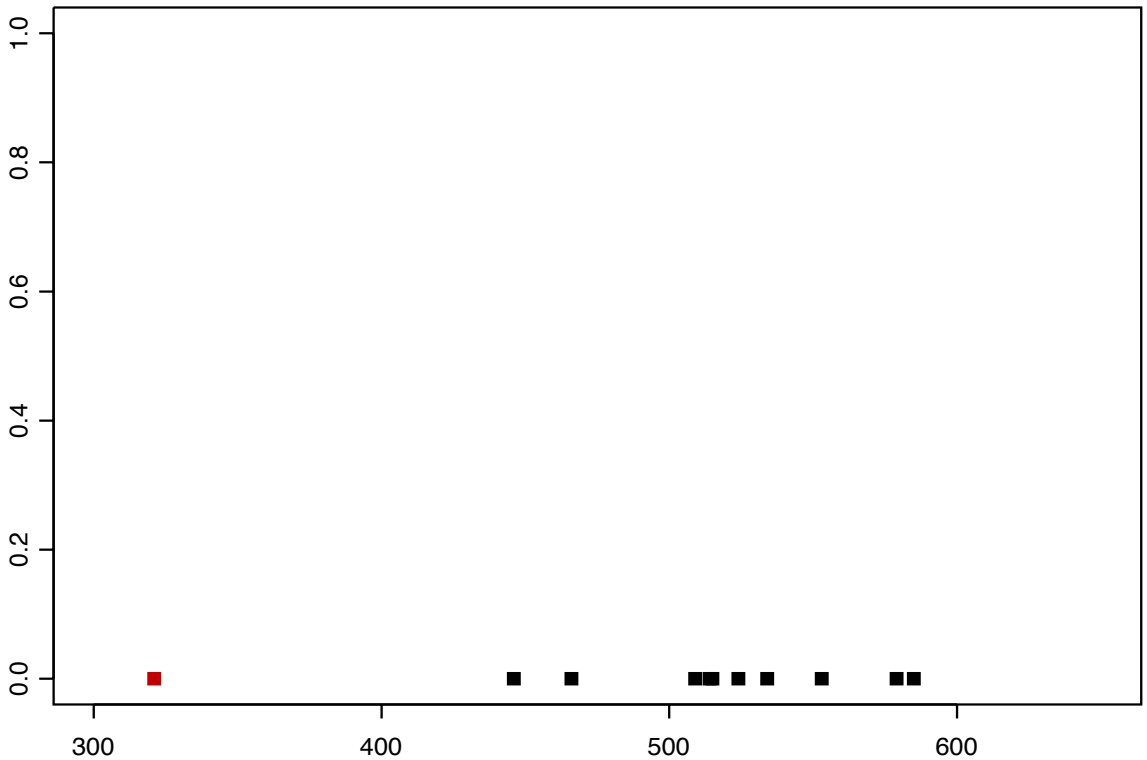
1 permutation



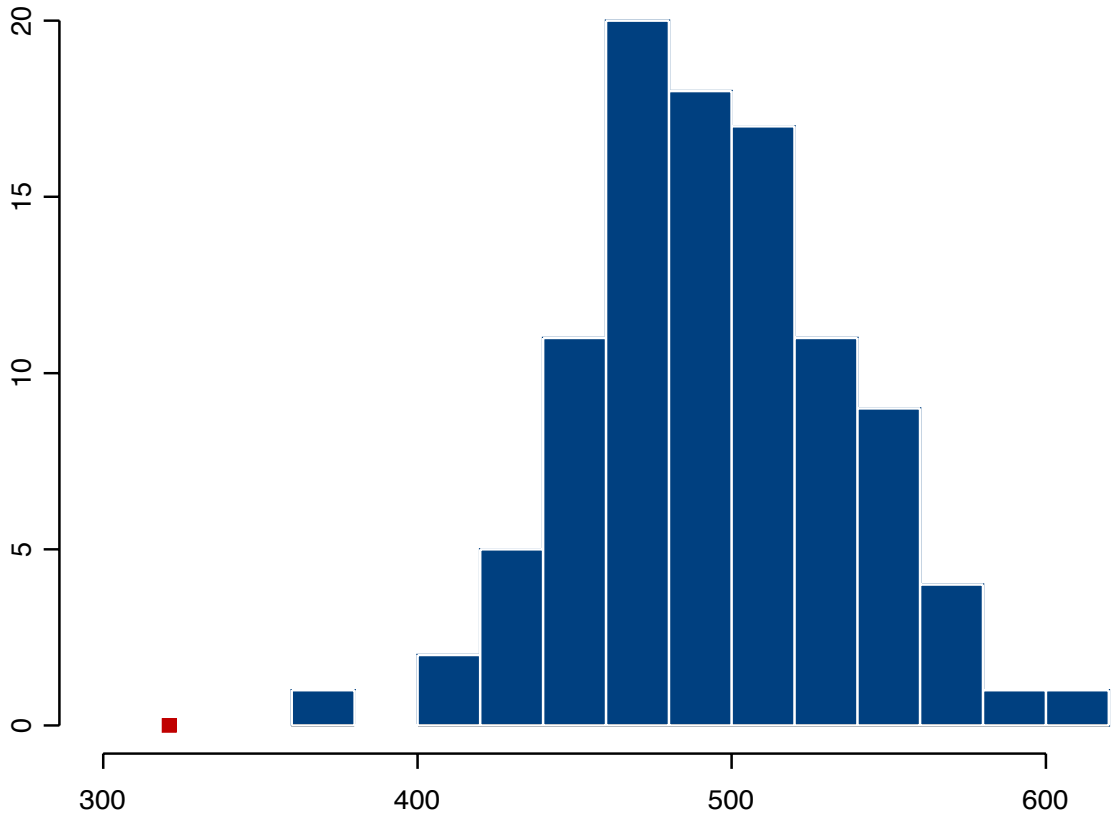
2 permutations



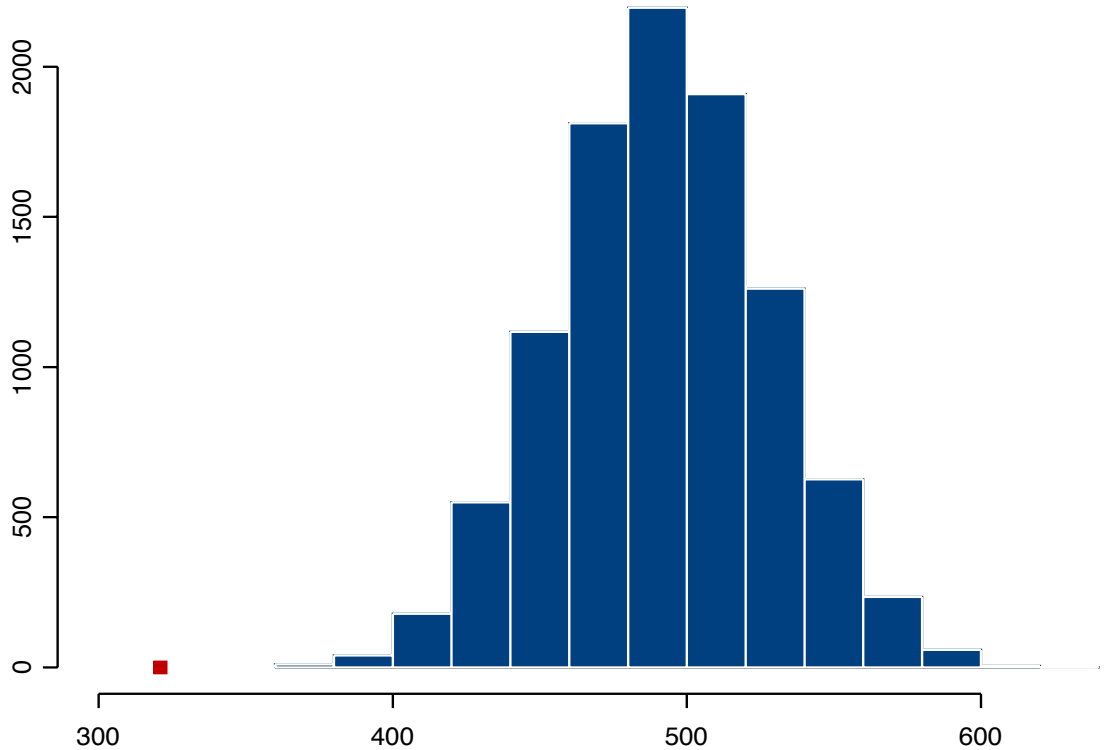
10 permutations



100 permutations



1000 permutations



Wilcoxon rank-sum test: p-value

- Based on observed distribution under null hypothesis, calculate probability of finding a more extreme result than the one observed
- Not a single permutation gives a more extreme result, hence $p < 1/1000$
- Such a test is called **permutation test** because distribution of test statistic under null hypothesis is obtained through permutations of the data
- Also possible to calculate exact distribution under null hypothesis

Analysis in R

```
wilcox.test(wbc ~ group, exact = T, data=rat)
```

```
    Wilcoxon rank sum exact test

data:  wbc by group
W = 21, p-value = 2.753e-07
alternative hypothesis: true location shift is not equal to
```

R subtracts $n(n+1)/2 = 300$ from the test statistic, hence the difference between the value that we computed ($W = 321$) and the value reported by R ($W = 21$)

Compare with t-test

```
t.test(wbc ~ group, data=rat)
```

```
    Welch Two Sample t-test

data:  wbc by group
t = -5.9107, df = 14.119, p-value = 3.667e-05
alternative hypothesis: true difference in means between gr
95 percent confidence interval:
 -33779.84 -15802.46
sample estimates:
mean in group 0 mean in group 1
       12215.92        37007.07
```

Wilcoxon rank-sum test

- Can be shown that the Wilcoxon rank-sum test actually tests if for 2 series of observations X and Y

$$P(X \geq Y) = \frac{1}{2}$$

- Consequently, a significant difference can be detected, even if center of distribution does not differ
- Similar nonparametric tests exist for related problems

Use of nonparametric tests

- Make **no assumptions** on distribution of data, hence suited when
  - data not normally distributed or sample too small to determine if normally distributed (alternative is to perform transformation)
  - there are outliers
- But **less sensitive/powerful** than parametric tests when these would be valid
  - although difference in power often not very large
- Not very flexible and don't easily give access to associated effect measures and confidence intervals

Overview tests for continuous outcomes

Parametric test (nonparametric alternative)

- Compare location to a number:
  - One-sample t-test (sign test)
- Compare location between 2 groups:
  - Paired groups: paired t-test (Wilcoxon signed-rank test)
  - Unpaired groups: unpaired t-test (Mann-Whitney test, Wilcoxon rank-sum test)
- Compare location between more than 2 groups:
  - One-way ANOVA (Kruskal-Wallis test)

Summary

- **Nonparametric tests** make no assumptions about the distribution of the data: useful for non-normal data, small samples or outliers
- **Wilcoxon rank-sum test**: replace observations by their ranks and compare rank sums between groups
- The distribution under H_0 can be obtained by permuting the data (**permutation test**)
- Nonparametric tests are somewhat less powerful than parametric tests when the latter are valid, and give less access to effect sizes and confidence intervals